

Muhammad Kandil

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Professional Summary

A highly motivated and detail-oriented Computer Science student with a robust foundation in programming, data structures, and software development. Proficient in programming languages such as Python, C#, and C++, with practical experience in machine learning, full-stack web development, databases, and cloud computing.

Education

- **October University For Modern Science And Arts**
Bachelor of Computer Science

Giza, Egypt
Graduation: Jun 2027

Project Experience

- **MASARI - CS Course Recommender & Interview Analyzer** | *Deep Minds 7th Edition Nominee* Giza, Egypt
 - **Full-Stack AI Engineering:** Engineered an intelligent educational companion using Python, FastAPI, and Vanilla JS, merging LLMs with dynamic data visualization dashboards.
 - **AI & Audio Integration:** Implemented generative technical grading with strict JSON LLM parsing and integrated WebRTC & Whisper API for real-time voice transcription during interactive mock interviews.
 - **Data Processing:** Developed a semantic search engine utilizing VADER sentiment analysis to accurately rank CS courses, powered by an asynchronous headless scraping engine.
- **FUNDR - Credit Risk Prediction & Interactive Simulator** | *DEPI R4 Graduation Project* Giza, Egypt
 - **Machine Learning Pipeline:** Developed a robust financial technology platform using an XGBoost classifier trained on 1.3 million historical loan records to accurately predict default probabilities.
 - **Model Interpretability:** Integrated SHAP tree explainers to provide transparent, mathematical breakdowns of risk factors for live applicant profiles.
 - **Generative AI Integration:** Connected the risk engine to an external LLM via the OpenRouter API to automatically author natural-language investment theses in milliseconds.
- **Forecastify** | *Machine Learning Course* Giza, Egypt
 - **Time-Series Forecasting:** Built a hierarchical sales forecasting system using the M5 Forecasting Accuracy dataset to accurately predict daily unit sales.
 - **Algorithm Evaluation:** Engineered and trained multiple predictive models, including Random Forest, XGBoost, LightGBM, Prophet, and Temporal Fusion Transformer (TFT).
 - **Performance Analytics:** Benchmarked and optimized models against rigorous statistical metrics including RMSE, MAE, R^2 , MAPE, and WAPE to ensure high forecasting accuracy.
- **DorMsa** | *Advanced Web Programming Course* Giza, Egypt
 - **System Architecture:** Designed a comprehensive university housing management system tailored for secure, seamless interactions between students, parents, and real estate brokers.
 - **Access Control:** Designed complex roles-based access control flows and verification protocols to ensure user data privacy and system integrity.
 - **Security Integration:** Integrated automated phone number verification utilizing One-Time Password (OTP) architectural patterns to establish trusted communication channels.
- **Togaaa Library** | *Mobile Device Programming Course* Giza, Egypt
 - **Full-Stack Integration:** Built a cross-platform book review app using Flutter and MongoDB, managing the complete lifecycle from UI design to backend data persistence.
 - **Database Architecture:** Designed a flexible NoSQL schema to efficiently store and retrieve complex book metadata and user-generated reviews.
 - **UX Optimization:** Implemented asynchronous data handling and state management to ensure a high-performance, responsive interface for real-time interactions.

- **Sudo-Cool** | *Analysis and Algorithms Course* Giza, Egypt
 - **Algorithm Engineering:** Developed an automated Sudoku solver utilizing a Recursive Backtracking algorithm to solve puzzles with 100% accuracy.
 - **Full-Stack Web App:** Built a web interface using Flask that allows users to input puzzles and visualize the solving process in real-time.
- **OS Visualizer** | *Operating System Course (2nd Place Winner)* Giza, Egypt
 - **Algorithm Simulation:** Developed a cross-platform tool to visually and interactively demonstrate Operating System algorithms, improving accessibility for students.
 - **Visual Architecture:** Designed and implemented a visual tool to simulate key operating system algorithms (e.g., CPU scheduling, memory management).
- **Super Market Management System** | *Data Structure Course* Giza, Egypt
 - **System Development:** Implemented core functionalities including inventory management, sales tracking, and customer management in C++.
 - **Data Structures:** Utilized advanced data structures such as arrays and linked lists for efficient data handling and high-speed retrieval.

Courses & Training

- **DEPI (Digital Egypt Pioneers Initiative) - Cohort 4**
 - **Data Science Course:** Intensive training in machine learning pipelines, data preprocessing, and predictive analytics.
- **NTI & Amazon Web Services**
 - **Cloud & Data Analytics:** Completed a rigorous 120-hour summer training program in Data Analytics. Completed AWS Introduction to Cloud 101, Networking, Storage, and Compute.

Technical Skills

Languages: Python, Java, C++, JavaScript, TypeScript, SQL, Dart

Frameworks & Tools: React.js, Next.js, Node.js, FastAPI, Flask, Flutter, Pandas, XGBoost

Databases: MySQL, MongoDB

Core Competencies: Machine Learning, Data Structures & Algorithms, Full-Stack Development, Cloud Computing, Technical Problem-Solving